

Karan Panwar

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Professional Summary

Applied AI Engineer and Researcher with hands-on experience developing LLM-powered applications, recommendation systems, NLP pipelines, and deep learning models. Led AI development at Klip Global, building systems that combine NLP, behavioural analytics, and NHS health data to deliver personalised AI solutions. First author of a peer-reviewed conference paper on scalable LLM-based digital mental health systems, published in Springer's *Artificial Intelligence in Healthcare (AiiH 2026)* proceedings. MSc Computer Science (Distinction) from Anglia Ruskin University, with research spanning quantum-classical hybrid neural networks and computer vision. Strong foundation in Data Science, including Python, SQL, data analysis, statistical modelling, machine learning, and data-driven problem solving. Seeking opportunities in Data Science, Machine Learning Engineering, or Applied AI where technical research can be translated into practical, measurable business impact.

Education

MSc Computer Science (Distinction)

Sep 2024 – Dec 2025

Anglia Ruskin University, Cambridge, UK

Key Modules: Machine Learning in Finance, AI for Imaging, Semantic Web Technologies, Advanced Web Solutions, Cybersecurity and AI

BSc Computer Applications (First Class)

May 2021 – May 2024

St. Xavier's College, Ahmedabad, India

Key Modules: Data Structures, Big Data Analytics, Database Management, Python Programming, Statistical Modelling

Publications

Panwar, K., Cirstea, S., Kalidindi, S., Chakraborty, A. (2026). *A Scalable LLM-Based System for Early Mental Health Intervention and Workforce Resilience*. In: *Artificial Intelligence in Healthcare (AiiH 2026)*, Lecture Notes in Computer Science, vol. 16877. Springer, Cham. DOI: 10.1007/978-3-032-35393-1_21

Core Technical Skills

ML & AI: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, Scikit-learn, XGBoost, LSTM, CNN, Generative AI, RAG, Prompt Engineering, Qiskit (QML)

Data & Analytics: Pandas, NumPy, SQL, Feature Engineering, Statistical Modelling, Power BI, Tableau, Matplotlib, Seaborn

Deployment & MLOps: FastAPI, Django, Flask, Docker, CI/CD Pipelines, Git, Streamlit

Web & Frontend: Next.js, React.js, TypeScript, RESTful APIs

Databases: MySQL, MongoDB, SQLite, Data Warehousing

Languages: Python, SQL, Java

Professional Experience

Research Assistant (Generative AI) – Klip Global

Sep 2025 – June 2026

London, UK (Remote from Cambridge)

- Architected **KAI Navigator**, an LLM-powered digital wellbeing platform integrating a hybrid recommendation engine, behavioural analytics, and NHS health data to deliver personalised, evidence-based guidance at scale.
- First-authored a peer-reviewed on this system, published in Springer's *Artificial Intelligence in Healthcare (AiiH 2026)* conference proceedings, covering the platform's Integrated Agent Architecture and hybrid recommendation model.
- Designed a hybrid recommendation engine combining Transformer-based NLP and clustering over an NHS-integrated dataset, improving guidance relevance over a baseline collaborative filter.
- Deployed a **LangChain + Hugging Face** conversational agent that improved user engagement and response quality scores in structured user testing sessions.

- Engineered a scalable Django backend integrated with a Next.js frontend, reducing API response latency and enabling production-ready deployment.
- Collaborated across product, data, and research teams to iterate on model performance, incorporating user feedback into weekly release cycles.

Research Assistant – Quantum Neural Networks

Jan 2025 – Apr 2025

Anglia Ruskin University, Cambridge, UK

- Designed Qiskit quantum circuits as drop-in replacements for classical CNN layers, improved user engagement and response quality scores on image classification benchmarks.
- Benchmarked quantum-classical hybrid models across multiple datasets, producing the strongest accuracy result in the study cohort.
- Contributed experimental design, benchmarking, and technical documentation for university research on quantum machine learning for imaging applications.

Selected Projects

KpAgent: AI-Powered Career Assistant

Next.js, TypeScript, NLP, Tailwind CSS

Built a full-stack career services platform with AI-driven resume analysis and personalised job recommendations. Engineered a custom interview preparation tool generating role-specific questions from user profiles and job descriptions. Delivered a responsive UI across all devices with sub-second interaction response times.

SmartStock: AI Financial Forecast Dashboard

Python, LSTM, FastAPI

Built a stock prediction dashboard using LSTM-based time-series models with real-time market API integration and financial sentiment analysis. Achieved forecasting accuracy competitive with published LSTM baselines on standard equity datasets.

EmotionSense: Real-Time Emotion Recognition

OpenCV, TensorFlow, Flask

Trained a CNN + MFCC model detecting facial and vocal emotions across seven emotion classes, recommending personalised media content via an adaptive algorithm. Achieved over 88% classification accuracy on held-out test data.

Cambridge Culture Portal

Django, HTML, CSS, SQLite

Led a four-member team building an interactive cultural platform showcasing Cambridge events and local community listings, delivering on time within a six-week sprint.

Certifications

AWS Certified Cloud Practitioner | IBM Data Science Professional Certificate | AI & ML using Python – IIT Madras via SWAYAM (Proctored, Score: 86%) | Quantum Computing Fundamentals – IBM Qiskit | Advanced Deep Learning – Coursera

Achievements & Leadership

- Graduated in the **top 10%** of MSc cohort with Distinction; awarded for academic excellence by faculty.
- First-authored a peer-reviewed conference paper published in Springer's Lecture Notes in Computer Science series (AIiH 2026).
- Selected by **Klip Global** to lead industrial AI research integrating LLMs with NHS health data analytics.
- Served as **Team Lead** across four cross-functional AI and web development projects at Anglia Ruskin University.